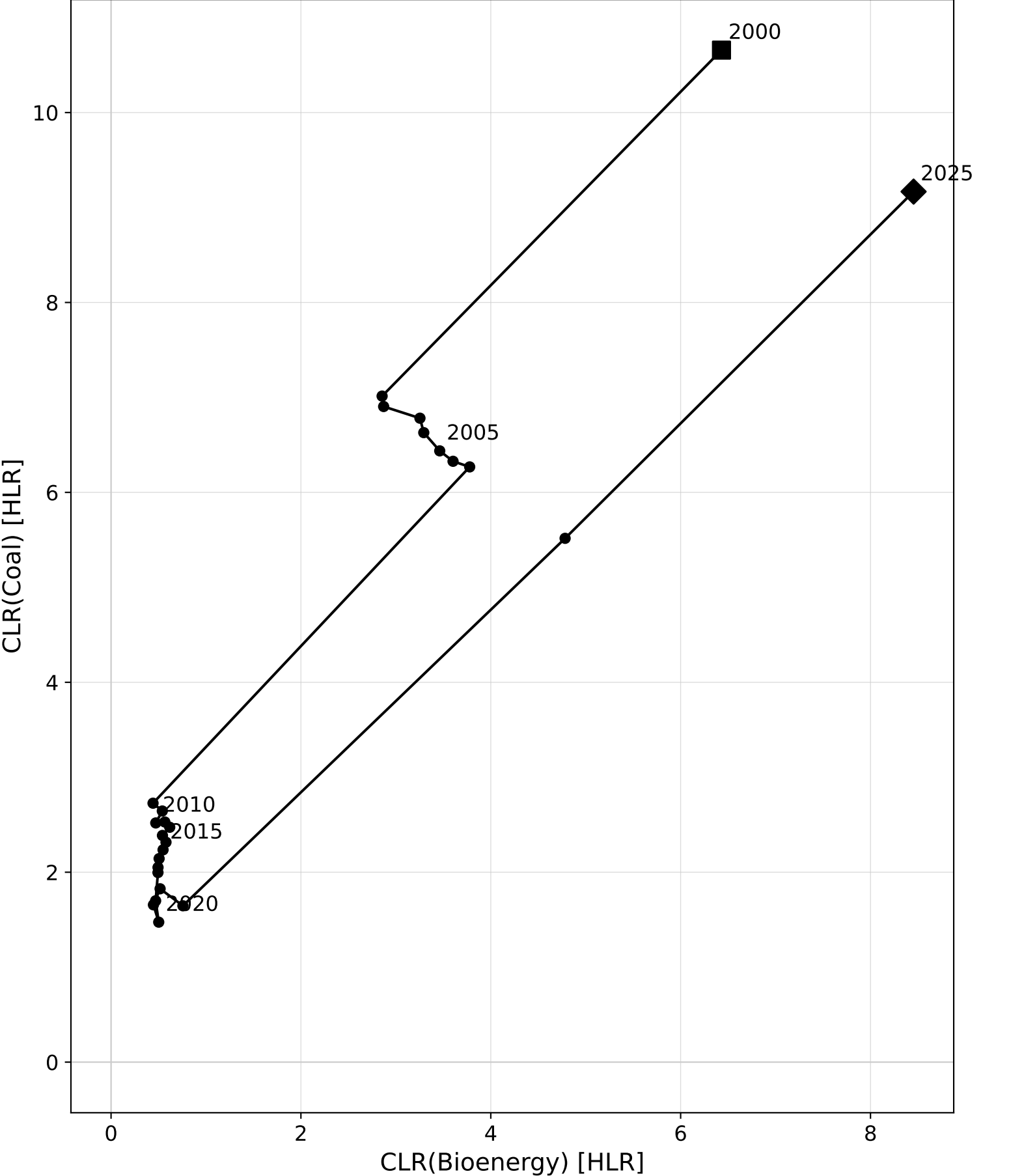


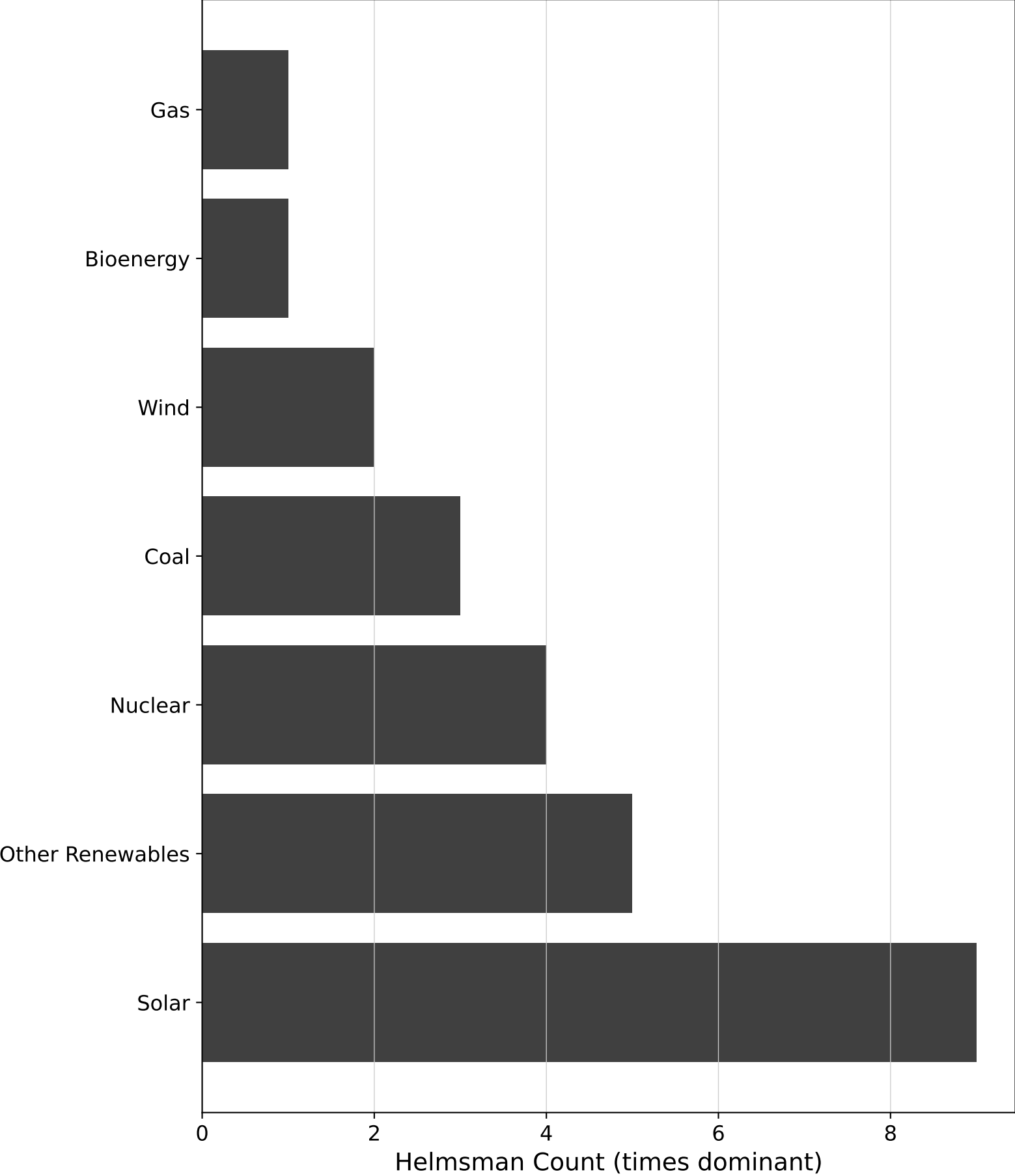
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



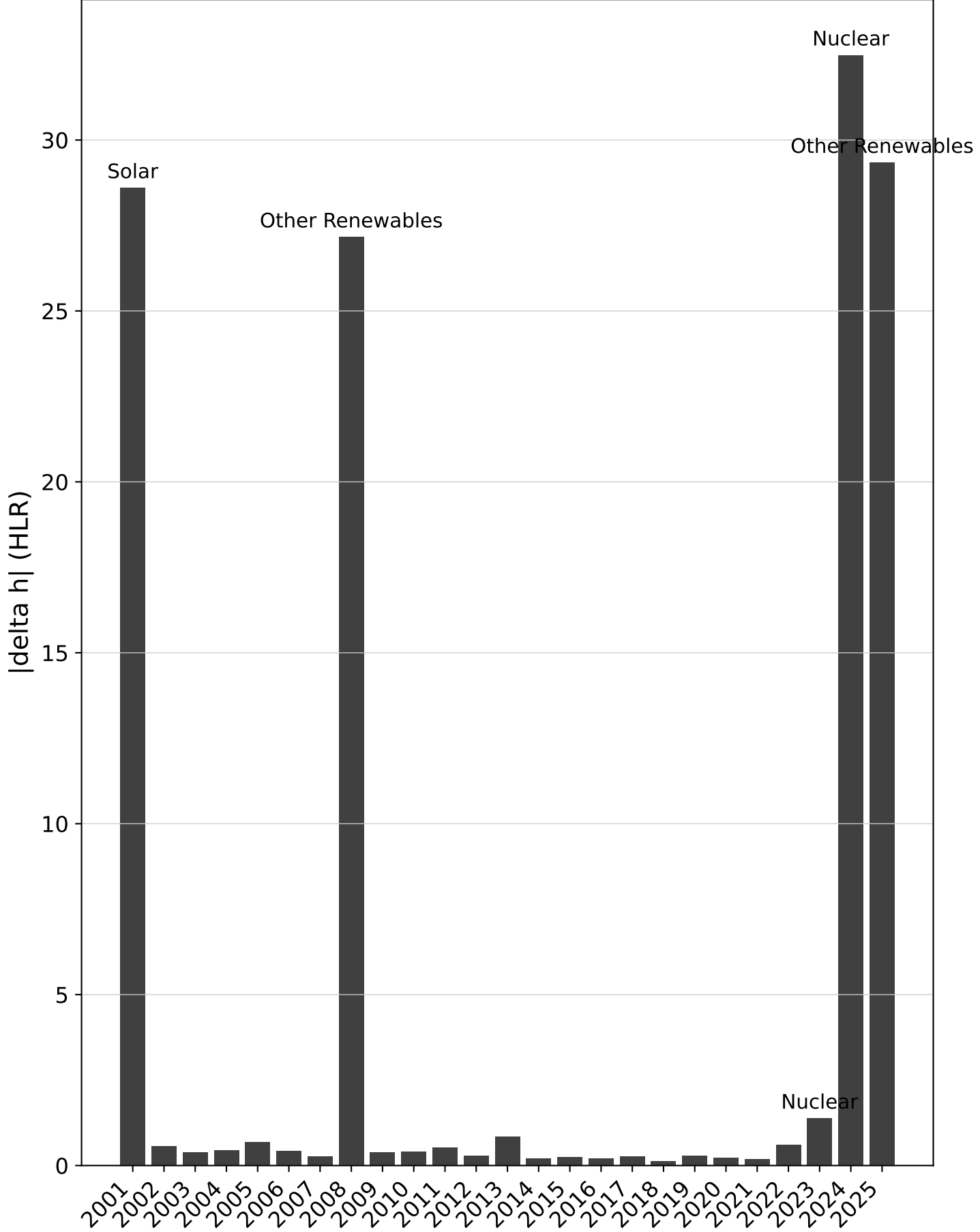
STAGE 2 – Aitchison Distance Matrix (all year pairs)


$$d_A(\text{HLR})$$

HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



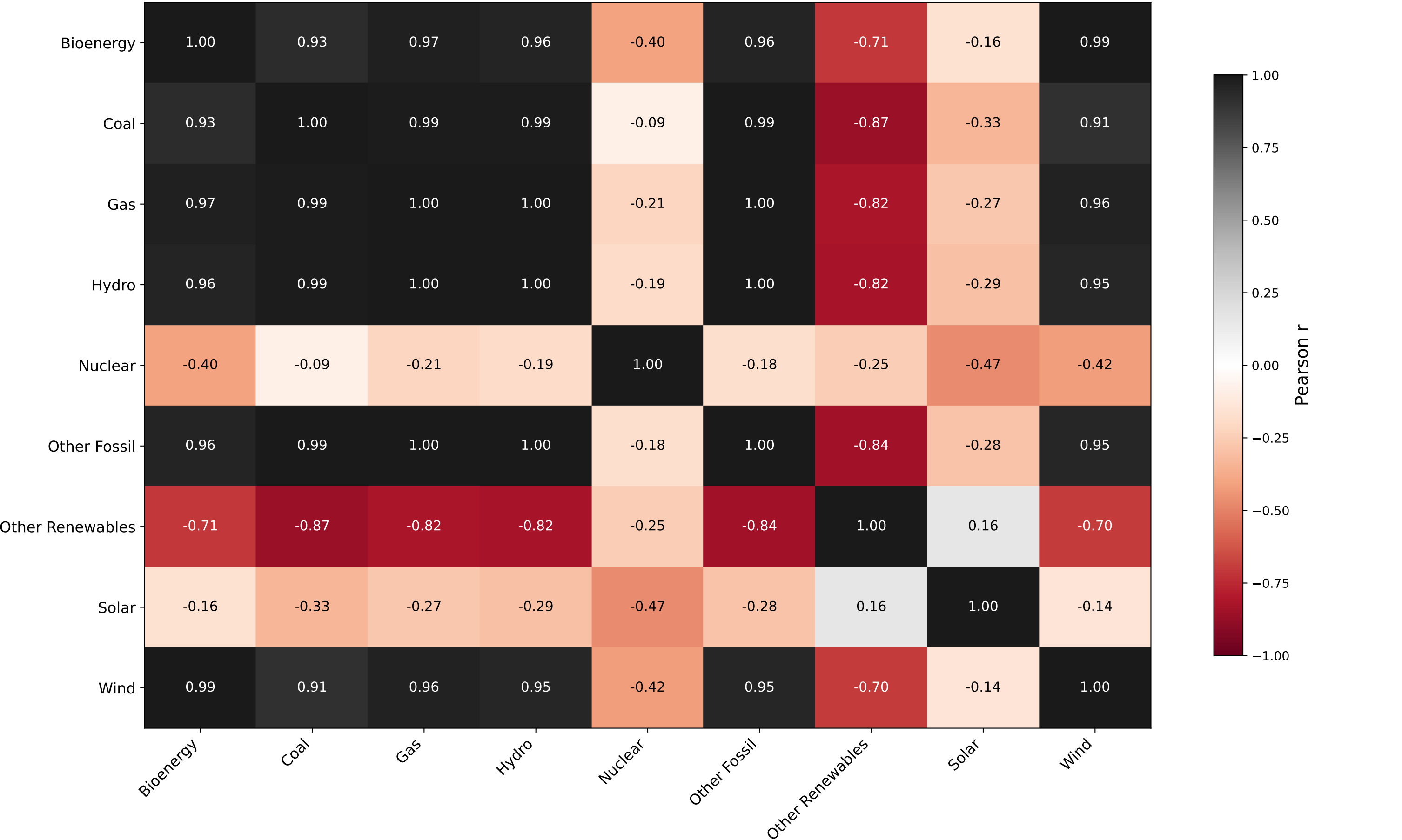
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2000	2024	63.5278	97.8	Nuclear
2	2000	2025	55.7717	71.3	Nuclear
3	2001	2024	52.1597	92.9	Nuclear
4	2002	2024	52.0311	92.5	Nuclear
5	2003	2024	51.9639	92.2	Nuclear
6	2004	2024	51.9001	92.0	Nuclear
7	2005	2024	51.7971	91.6	Nuclear
8	2006	2024	51.7650	91.4	Nuclear
9	2007	2024	51.5963	91.0	Nuclear
10	2021	2025	45.0284	52.8	Nuclear
11	2016	2025	45.0028	54.7	Nuclear
12	2019	2025	44.9708	53.4	Nuclear
13	2015	2025	44.9648	55.5	Nuclear
14	2020	2025	44.9643	52.2	Nuclear
15	2014	2025	44.9451	56.2	Nuclear

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2000	2021	2025	970.4208	1.26
2	2000	2020	2025	969.6442	1.26
3	2000	2019	2025	965.7014	1.27
4	2000	2022	2025	964.6295	1.26
5	2000	2018	2025	963.3611	1.27
6	2000	2016	2025	961.1902	1.28
7	2000	2015	2025	958.6905	1.28
8	2000	2017	2025	958.0769	1.28
9	2000	2014	2025	955.5446	1.28
10	2000	2023	2025	951.2340	1.29

STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.4307	Hs-3	1575.1	0.00	0.0000	---	1000000000000000
2001	0.4142	Hs-3	897.4	39.61	30.3382	Solar	1000000000000000
2002	0.4057	Hs-3	897.5	1.17	0.7326	Solar	1000000000000000
2003	0.3892	Hs-3	899.7	1.05	0.6614	Bioenergy	1000000000000000
2004	0.3678	Hs-3	902.1	0.91	0.5772	Solar	1000000000000000
2005	0.3411	Hs-3	905.0	1.28	0.8104	Solar	1000000000000000
2006	0.3261	Hs-3	907.1	0.82	0.5187	Solar	1000000000000000
2007	0.3068	Hs-3	910.1	0.72	0.4555	Nuclear	1000000000000000
2008	0.2851	Hs-2	63.6	19.49	28.8119	Other Renewables	31614
2009	0.2725	Hs-2	61.5	3.19	0.4604	Solar	29398
2010	0.2572	Hs-2	56.0	3.16	0.5558	Solar	20807
2011	0.2405	Hs-2	60.6	5.40	0.7791	Solar	30250
2012	0.2326	Hs-2	54.9	2.44	0.4902	Other Renewables	20660
2013	0.2353	Hs-2	42.7	2.45	0.9251	Other Renewables	7871
2014	0.2245	Hs-2	40.1	2.51	0.3463	Other Renewables	6188
2015	0.2119	Hs-2	37.8	2.92	0.3706	Wind	4917
2016	0.1987	Hs-2	35.0	2.70	0.3670	Gas	3774
2017	0.1842	Hs-2	35.8	3.02	0.3220	Wind	4028
2018	0.1785	Hs-2	34.6	2.15	0.2455	Solar	3513
2019	0.1489	Hs-1	32.9	3.34	0.3675	Coal	3000
2020	0.1372	Hs-1	31.0	3.00	0.3438	Coal	2464
2021	0.1426	Hs-1	30.6	2.79	0.2716	Coal	2412
2022	0.1682	Hs-2	31.9	7.31	0.7217	Nuclear	2699
2023	0.1773	Hs-2	33.6	15.33	1.5336	Nuclear	2667
2024	0.1907	Hs-2	923.3	68.52	34.4578	Nuclear	1000000000000000
2025	0.1958	Hs-2	1633.3	40.25	31.1242	Other Renewables	1000000000000000